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Practical Performa

Academic Year	:	2025-26	Semester	:	3 rd
Course code	:	CSUC201	Course name	:	Fundamentals of Data Structure and Algorithms

Practical- 8 TREES AND BINARY SEARCH TREES

Questions and Answers (Handwritten):

8.1 Match the Following (Tree Concepts)

Match the tree types in **List-I** with their corresponding **properties/applications** in **List-II**:

List – I (Tree Concept) | List – II (Property / Application)

- (a) Binary Search Tree (BST)
- (b) AVL Tree
- (c) Red-Black Tree
- (d) B-Tree
- (e) Binary Heap

- (i) Used in implementing **priority queues** (min/max).
- (ii) Balanced binary tree where **rotations** keep the balance factor in $\{-1, 0, 1\}$.
- (iii) Multi-way search tree widely used in **databases and file systems**.
- (iv) Keys in left subtree $<$ node key $<$ right subtree.
- (v) Maintains balance using **color property**; guarantees $O(\log n)$ search, insert, delete.

Answer Codes:

- (a) → ____
- (b) → ____
- (c) → ____
- (d) → ____
- (e) → ____

8.2 (AVL Balancing)

We use an AVL tree (rotate to keep balance factor in $\{-1,0,1\}$). Insert the following keys in given order into an empty AVL tree:

30, 20, 40, 10, 25, 35, 50, 5, 15, 27

For this sequence:

- (a) Insert the keys step by step. For every insertion that causes imbalance show the insertion path (nodes visited from root to insertion point) and list the rotation(s) applied (LL / RR / LR / RL), including which node(s) the rotation is performed on.
- (b) Show the AVL tree after all insertions. You may draw it or give a parenthesized representation (node(key, left-subtree, right-subtree)). Also list the balance factor for every node.
- (c) What is the height of the AVL tree after all insertions? (Assume height of a single node = 1.)
- (d) How many rotations in total were performed? Break down into single (LL/RR) and double (LR/RL) rotations.
- (e) Policy check (optional): If there is a policy to rebuild the entire tree into a perfectly balanced BST whenever more than 3 rotations have been performed during a sequence of insertions, would that policy trigger here? If yes, provide the rebuilt perfectly balanced BST (inorder of current keys then rebuild) and its height.

8.3 (Binary Search Tree Traversal)

Write an algorithm (in pseudocode or any programming language you prefer) to **print all nodes of a Binary Search Tree (BST) in ascending order**.

Instructions:

1. You are given a BST where each node contains a key, left child, and right child.
2. Use the **Inorder Traversal** strategy to print nodes.
3. Do **not** use recursion. Instead, use an **explicit stack** (or list) to simulate recursion.

50

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30 70

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20 40 60 80

Grade/Marks
Date

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